

System Background: The MEDIX System Integrated Modules of Care

Patient Information Management

◆ *Unified Patient Dashboard*

- Complete patient overview at a glance
- Real-time appointment tracking with visual status indicators
- Timeline showing appointment history.
- Instant access to patient statistics and care metrics

◆ *Comprehensive Patient Profiles*

- Complete demographic information (personal details, contact information, emergency contacts)
- Medical history with chronological tracking
- Current conditions and allergies

Appointment

Feature

◆ *Dual-Booking*

- Patient Self-Service Booking
- Nurse-Assisted Booking

◆ *Automated Triage*

1. Patients enter their reason for booking in plain language
2. Modified Dijkstra algorithm with fuzzy matching processes the input against our keyword database
3. Automatic extraction of medical indicators and symptoms

◆ *Dynamic Priority Assignment*

1. **Emergency:** Critical conditions requiring immediate attention
2. **Urgent:** Time-sensitive health concerns
3. **Normal:** Routine care and check-ups

◆ *Smart Department & Doctor Matching*

- Automatic department suggestion based on matched symptoms.
- Smart load balancing with priority queue assigns patients to doctors with optimal availability.

Scheduling

Feature

◆ *Real-Time Availability*

- Dynamic calendar displays each doctor's unique work schedule
- Available days appear clickable (white), unavailable days are hidden
- Instant synchronization across all booking interfaces

◆ *Interactive Timeslot Selection*

- Visual availability indicators: Green (available) vs Red (booked)
- Real-time updates prevent double-booking
- Optimized time management for healthcare providers

◆ *Manual Confirmation of Appointment*

- Doctor must manually approve an appointment to make it official.
- Automatic email notifications upon scheduling confirmation
- Patient are sent different email reminders depending on the status of their appointment. (Cancelled, Pending, Scheduled, or Completed)

System Background: The Foundation of Integrated Care

Referral Management

◆ **Referrals Sent Tab**

1. Track all sent referrals
2. Monitor referral status and follow-up requirements
3. Maintain comprehensive referral history for each patient

◆ **Referrals Received Tab**

1. Track all received referrals
2. View detailed referral details

◆ Referral Creation Quick Link

1. Doctors must manually put in all the required patient details such as:
 - a. Patient diagnosis and medical history
 - b. Current symptoms and medication lists
 - c. Known allergies and clinical notes

Billing

◆ Automated Billing Calculations

- Automatic computation of medical charges and fees. (e.g. discounts & final fees),

◆ Centralized Billing Records

- Patients can see their own billing records on the sidebar
- Admins can see all registered patients billing records.

◆ Payment Gateway Integration

- The payment gateway workflow is handled by Paymongo.
- Demo transaction capability using PayMongo test cards

Doctor/Staff Management

◆ User Management Dashboard

- Admins can add or remove staff/doctors registered in the system.
- Admins can view the workdays / schedules of doctors along with their profile information.
- Admins can view information about the Staffs (Nurses) in the system.
- Admins can check every member designated department.
- Each Management Table is separated in the Sidebar.

System Administration / Security

◆ Admin Management Dashboard

1. Management of all registered users in the system.
2. Adding and Removing of Doctors and Staff Members
3. Accessing and Managing Doctor Profiles and their respective appointments.
4. Viewing of all types of personal information.
5. Adding services to the billing module which handles the prices of service within a hospital.
6. Managing of billing operations including initial and final bills.

7. Adding of patient vital signs, accessing medical records, diagnosis history and appointment data for monitoring.
8. Adding Keywords on the Keyword Manager which is then stored in the database to be used for the Appointment Algorithm.
9. Has Access to Sentry, the application performance and application platform installed in the System.

Our Integrated Hospital Management System (HMS) has six features.

Bonus System Features

- It's **80 percent mobile responsive**. It can support smaller devices ranging from the iPhone SE to Samsung Galaxy A51
- We used **Neon Serverless Postgres** database which can **dynamically** scale up computing power during periods of high activity and down to zero when idle. (Zero waste) (\$20)
- The system is already packaged with **Sentry, a powerful open-source application monitoring and error tracking tool** designed to help developers detect, diagnose, and resolve issues in real-time across web and mobile applications.
- It has a **comprehensive search feature** particularly only found on the appointment tab. It uses keywords to search specific information.
- The payment processing is handled by Paymongo, although you can only do test transactions for now. In its full functionality, it supports payments of **Visa/Mastercard debit and credit cards (straight), BillEase (Buy Now, Pay Later), GCash, GrabPay, Maya, and Online banking via BPI/UBP.**